

```
Program on switch as an expression :
-----
//switch as an expression
//WAP to verify whether the given day is weekday OR weekend
void main()
{
    int choice = Integer.parseInt(IO.readLine("Enter your choice 1 - 7 :"));

    String result = switch(choice)
    {
        case 1,2,3,4,5 -> "Weekday";
        case 6,7 -> "Weekend";
        default -> "Invalid Day";
    };
    IO.println("Today is, Day "+choice+" so, It is a "+result);
}

SwitchDemo6.java
-----
/* Employee performer of the year
Grade A -> High-level performer [bonus amount is 20% of the salary]
Grade B -> Mid-level performer [bonus amount is 15% of the salary]
Grade C -> Basic level performer [bonus amount is 10% of the salary]
No grade -> 5 % of the salary
*/
void main()
{
    char grade = IO.readLine("Enter employee grade :").toUpperCase().charAt(0);
    double salary = Double.parseDouble(IO.readLine("Enter employee salary :"));

    double bonus = switch(grade)
    {
        case 'A' ->
        {
            IO.println("High-level performer");
            yield salary * 0.20;
        }
        case 'B' ->
        {
            IO.println("Mid-level performer");
            yield salary * 0.15;
        }
        case 'C' ->
        {
            IO.println("Basic-level performer");
            yield salary * 0.10;
        }

        default -> salary * 0.05;
    };

    IO.println("Employee Salary amount is : "+salary);
    IO.println("Employee bouns amount is : "+bonus);
    IO.println("Employee total salary is : "+(salary + bonus));
}

SwitchDemo7.java
-----
void main()
{
    long x = 12; //long, float, double and boolean are not allowed

    switch(x)
    {
        case 12L:
    }
}

SwitchDemo8.java
-----
void main()
{
    int x = Integer.parseInt(IO.readLine("Enter the value of x :"));
    final int y = 10;

    switch(x)
    {
        case y -> IO.println("It is case 10");

        case 5 + 6 -> IO.println("It is case 11");
    }
}

SwitchDemo9.java
-----
While working with byte, short and char the label of the case must be within the range only otherwise we will get compilation error

void main()
{
    short b = 120;

    switch(b)
    {
        case 120 -> IO.println("Case 120");
        case 127 -> IO.println("Case 127");
        case 128 -> IO.println("Case 127");
        case 33000 -> IO.println("Case 127"); //error out of the range
    }
}
}
```

2.Iteration OR Looping Statements :

Loops in Java:

If we want to execute **the same code** multiple times then writing the same code again and again is not a good choice.

Example:
I want to print "Welcome to Java" 10 times then writing this print statement 10 times is not a good programming practice.

```
Repeat.java
-----
//Not a good choice
void main()
{
    IO.println("Welcome to Java");
    IO.println("Welcome to Java");
    IO.println("Welcome to Java");
    IO.println("Welcome to Java");
    IO.println("Welcome to Java");
    IO.println("Welcome to Java");
    IO.println("Welcome to Java");
    IO.println("Welcome to Java");
    IO.println("Welcome to Java");
    IO.println("Welcome to Java");
}
```

Here we should use looping statement:

Loops in Java are used to execute a **block of code multiple times** as long as the given Condition is **true**. It also comes under control flow statement.

It helps us to reduce code duplication as well as to write the logic **once** and control how many times it runs using a condition.

Types of Loops in Java:
Java provides four types of Loops:

- 1) do while loop
- 2) while loop
- 3) for loop
- 4) for-each loop

Common Characteristics of loop:

a) Initialization (Starting point of the loop)
b) Condition (Ending point of the loop)
c) Updating (Increment/Decrement)

do-while loop

It is an **exit-controlled loop** because without checking the condition, it will execute the body at-least once.

```
Syntax:
do
{
    // statements
}
while(condition);

Programs :
-----
DoWhile1.java
-----
void main()
{
    do
    {
        int x = 1; //x is local variable but accessible within the do
        IO.println(x); //block only
        x++;
    }
    while (x<=10); //error
}

DoWhile2.java
-----
void main()
{
    int x = 1;
    do
    {
        IO.println(x);
        x++;
    }
    while (x<=10);
}

DoWhile3.java
-----
void main()
{
    int x = 1000;
    do
    {
        IO.println(x); //1000 exit control behaviour
        x++;
    }
    while (x<=10);
}

DoWhileDemo4.java
-----
void main()
{
    int x = 0; //0
    do
    {
        IO.println(x); //0 0
        x += x++; //x = x + x++; //x = 0 + 0
    }
    while (x<=5);
}

Infinite loop, Output is 0

DoWhileDemo5.java
-----
void main()
{
    int x = 1; //6
    do
    {
        IO.println(x); //1 2 4
        x += x++; // x = x + x++; x = 3 + 3
    }
    while (x<=5);
}

void main()
{
    int i = 1;

    do
    {
        IO.println(i);
    }
    while (i++ <= 3);
}
```

```
-----
void main()
{
    int i = 1;

    do;
    while (i++ < 5);

    IO.println(i);
}

-----
void main()
{
    int i = 1;

    do
    {
        IO.println(i);
    }
    while ((i = i + 1) < 5);
}
```

```
-----
void main()
{
    int i = 1;

    do
    {
        IO.print(i++ + " ");
    }
    while (i++ < 5);

    IO.println("\ni = " + i);
}
```

2) while loop:

It is an **entry-controlled loop** because first, it will verify the **condition**, If Condition is **true** then only body will be executed.

It is preferable when number of iterations are **not fixed**.

```
Syntax:
while(condition)
{
    // statements
}

While1.java
-----
//Print 0 to -10
void main()
{
    int x = 0;

    while(x >= -10)
    {
        IO.print(x+"\t");
        x--;
    }
}

While2.java
-----
void main()
{
    while(false) //Unreachable
    {
        IO.println("Hello learner!!");
    }

    IO.println("Have you enjoyed it!!");
}

While3.java
-----
void main()
{
    boolean isValid = false;

    while(isValid)
    {
        IO.println("Hello learner!!");
    }

    IO.println("Have you enjoyed it!!");
}

While4.java
-----
void main()
{
    while(true)
    {
        IO.println("Hello learner!!");
    }

    IO.println("Have you enjoyed it!!"); //Unreachable
}

While5.java
-----
void main()
{
    final int x = 10;
    final int y = 20;

    while(x>y)
    {
        IO.println("x is greater than y");
    }

    IO.println("Out of the while loop");
}
```